NFTs – Not Frivolous Tech

A web search for 'non-fungible token' or NFT throws back endless news about limited-edition virtual sneakers, digital art, memes, tweets, and trading cards being sold for unrealistic prices. But NFT means much more. It is serious tech with extremely great potential in the digital and physical worlds. Here is what you need to know about NFTs and their applications.

A non-fungible token (NFT) of Chris Torres' *Nyan Cat*, a video of a flying cat with a pop tart body leaving behind a rainbow trail, sold for $580,000, while *Everyday: The First 5000 Days*, a collage of 5000 digital images by Mike Winkelmann, sold for $69 million. A 21-year-old Dutch student recently put up his soul for sale as an NFT, and at the time of writing this article its price had crossed $3.4 million. Closer home, four Thar based NFTs auctioned by Mahindra & Mahindra in March 2022 sold for a sum of ₹26,00,000!

Seeing these examples, one may be fooled into thinking non-fungible tokens or NFTs are all about people burning cash in collectibles and memorabilia. However, NFT as a technology means much more, and can potentially be used for many more real applications that can affect our daily lives, including medical record maintenance, improving supply chain efficiency, industrial design prototyping, and protecting intellectual property (IP). It can be a very effective way of digitising real-world objects and processes.

Figure 1: Mahindra's Thar based NFT - Standing Tall (Source: auto.mahindra.com)

Let us take a quick look at what NFTs are, a deeper look at their role in the art, collectibles and memorabilia space, and about other uses they can be applied to.

What is an NFT?

A non-fungible token or NFT is a unit of data stored on a digital ledger, such as the blockchain. It is also a crypto-token like digital currency, but here the unit of data is associated with a particular digital or physical asset—say, a file, a work of art, a degree certificate. The data is used to give important information about the asset, such as the creator's signature, current ownership, history, and terms of use or license.

Let us take a look at its characteristics.

It is non-fungible: Cryptocurrencies like Bitcoin or Ether are fungible tokens. That is, if you take two Bitcoins, they represent the exact same value, which makes them interchangeable or fungible. On the other hand, NFTs represent assets that are rare and unique, and not replaceable by one another, which makes them non-fungible.

It is tradeable: You can sell or trade NFTs in digital marketplaces, such as OpenSea, Foundation, Sorare, or Decentraland, using supported digital currencies stored in digital wallets.

It is immutable: Once an NFT has been encoded using blockchain technology, you cannot alter the